

An Exploration of Loss Functions in Multimodal Models and Their Impact on Downstream Performance

Master thesis by Philipp Scholl

Outline

1. Introduction (Motivation & Research aim)
2. Study design (Architecture, Implementation, Loss functions, Evaluation methods)
3. Related work
4. Results
5. Discussion
6. Limitations and Future Work

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Motivation

- **Multimodal Learning:** learning unified representations from different data modalities
 - Focus of thesis: Vision-and-Language learning
- **Multimodal models differ across multiple components:**
 - Architecture: Single-stream vs. Dual-stream
 - Loss functions
 - Training data, specific architectural components, etc.
- Hard to **determine the impact of individual components** on model performance
- **Loss functions determine what the model learns from data**, impacting representation quality and performance on downstream tasks

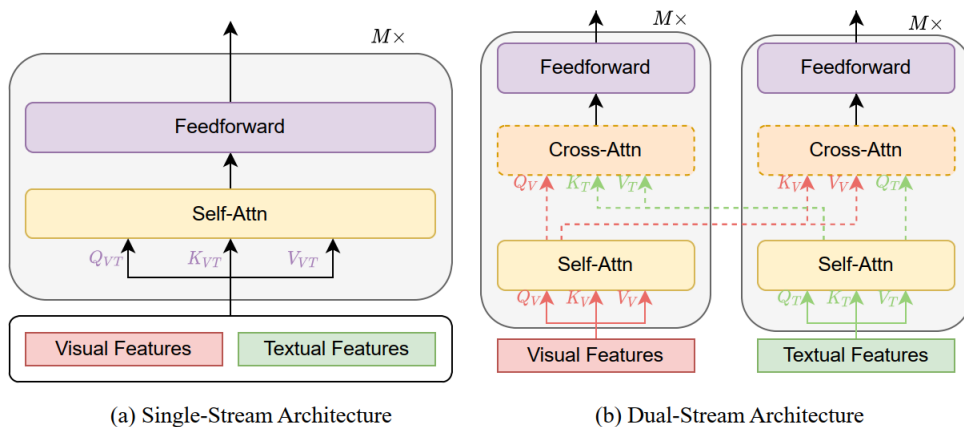


Fig.: Single-stream vs Dual-stream architecture

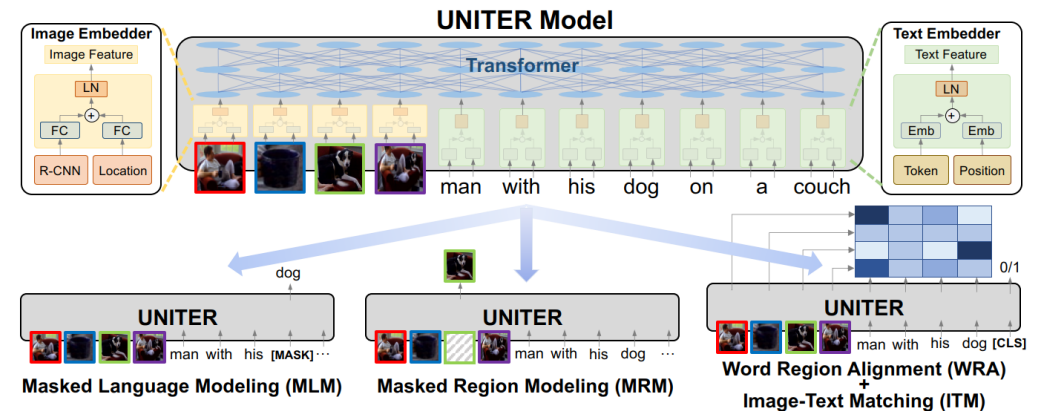


Fig.: Single-stream model “UNITER”, trained with four different loss functions

Research aim

- Question: what is the **specific influence of a loss function** on a model's *representation quality and downstream task performance*?
- Approach: *fix all other components* and parameters and only **ablate the choice of loss function**
- Construct a model architecture that can accommodate *a combination of four loss functions*:
 - Multi-modal: **Contrastive Image-Text (CLIP)** and **Image-Text Matching (ITM)**
 - Uni-modal: **Contrastive Image-only (SimCLR)** and **Text-only Masking (MLM)**
- Experiments:
 - **Test models on three downstream tasks**: Zero-Shot Image Classification, Zero-Shot Image-Text Retrieval, Linear probe Image Classification
 - **Ablation study on size of training dataset**
 - **Analyze similarity of representations with *Centered Kernel Alignment***

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Model architecture

- **Dual-stream** architecture
 - Image: Vision Transformer Base
 - Text: BERT Base
- **Modalities aligned at projection layer**
 - Contrastive Image-Text loss
 - Image-Text Matching loss
- **No cross-attention layers**
 - CLIP excludes cross-modal interactions!
- **Unimodal losses applied to stream**
 - *No cross-modal conditioning!*
- **All losses are weighted equally**

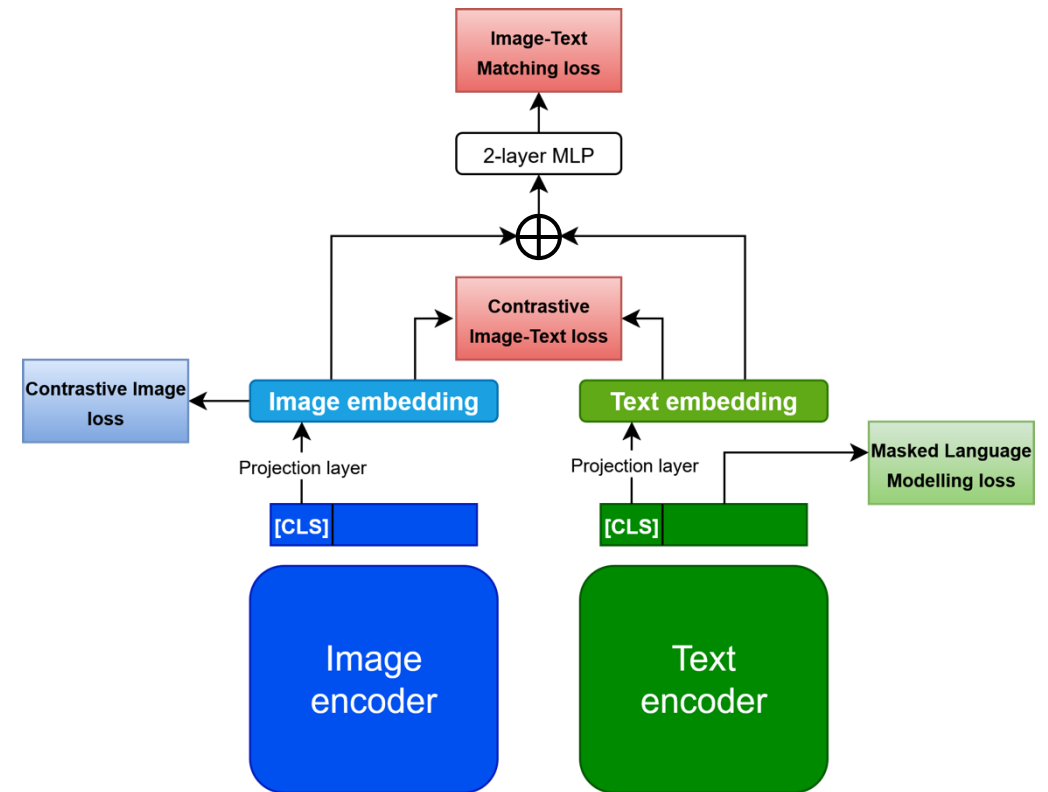


Fig.: Diagram of model architecture

Each encoder:

- 12 layers: dim=768
- Projection layer: dim=512

Implementation details

- Training dataset: **3.95M** image-caption pairs
 - *Conceptual Captions 3M* 3.3M examples
 - *MS COCO Captions* 590K examples
 - *Visual Genome (LLaVA split)* 60K examples
- Training from scratch (no pre-trained weights)
- Loss combinations: **12 models in total**
 - **8 CLIP models:** CLIP + ITM/SimCLR/MLM
 - **4 Non-CLIP models:** SimCLR, SimCLR + ITM, SimCLR + ITM + MLM, MLM + ITM
- Dataset ablation: **5%, 10%, 20%, 40%, 60%, 80%** fractions

Table 3.6: Number of training samples used per dataset fraction

Fraction (%)	Number of Samples
5	197,500
10	395,000
20	790,000
40	1,580,000
60	2,370,000
80	3,160,000
100	3,950,000

Loss functions: Contrastive Image-Text (CLIP) and Contrastive Image-only (SimCLR)

Multimodal loss: CLIP

Contrast matching image-text pairs with non-matching image-text pairs in mini-batch, maximizing similarity of the former while minimizing similarity of the latter

(1) Contrastive pre-training

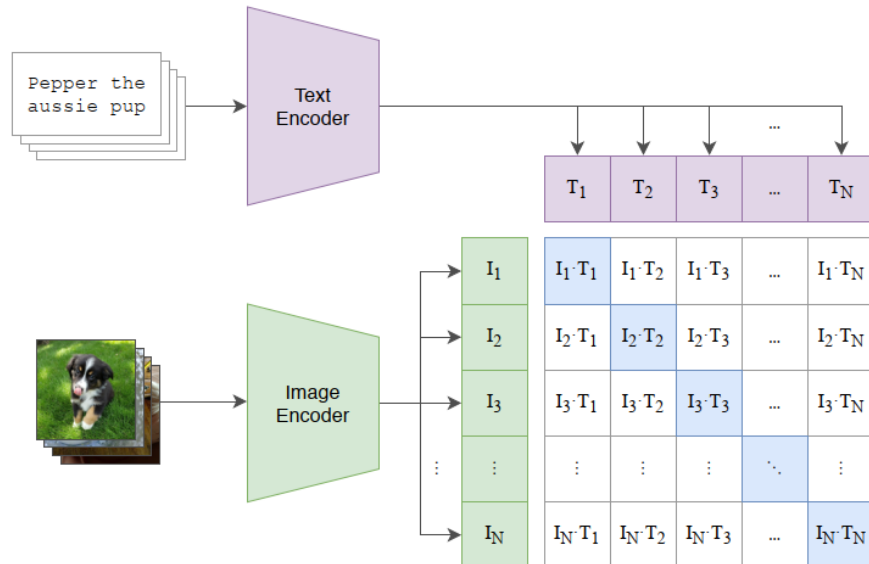


Fig.: Contrastive Image-Text pre-training

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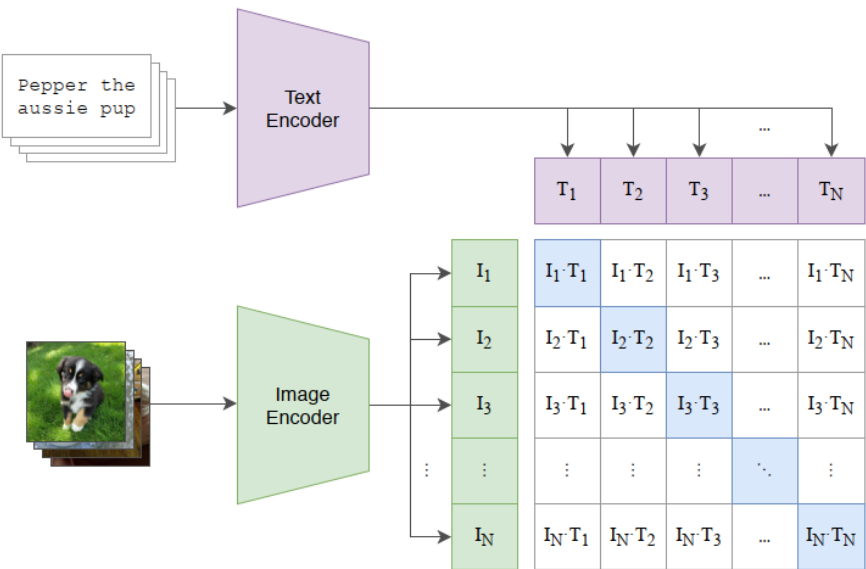


Fig.: Contrastive Image-Text pre-training

Unimodal image loss: SimCLR

Contrast pairs of two different transformations of same image with non-matching pairs in mini-batch, maximizing similarity of the former while minimizing similarity of the latter

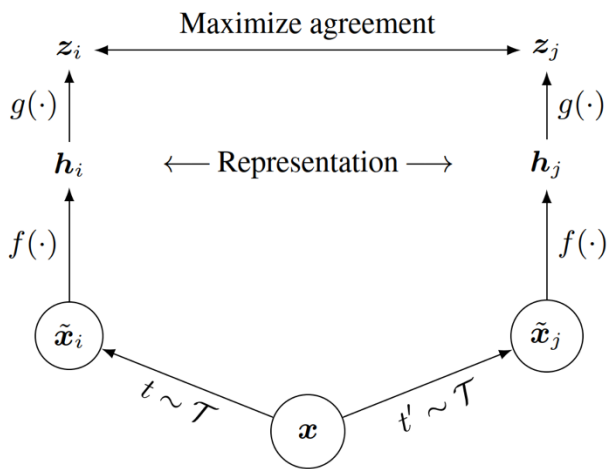
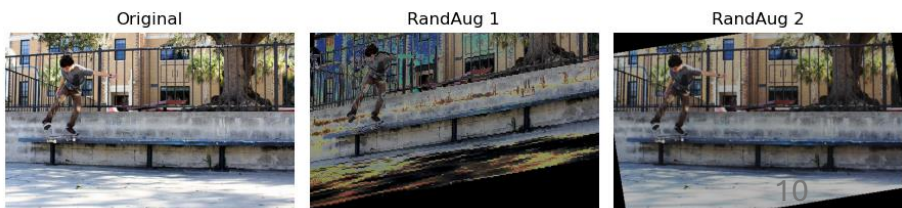


Fig.: Diagram of SimCLR pre-training

Fig.:
RandAugment
image
transformation



Loss functions: Image-Text Matching (ITM) and Masked Language Modelling (MLM)

Multimodal loss: ITM

Predict if an image-text pair matches or not

- **Concatenate image and text embeddings** from projection layer
- Feed through **2-layer MLP** to obtain **two-class probability “matching/not matching”**

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Multimodal loss: ITM

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Unimodal text loss: MLM

Predict tokens that were masked at input

- Encodes bidirectional sentence context in tokens

Text-only masking loss

- *Not conditioned on visual features!*
- CLIP loss excludes cross-modal loss functions

Evaluation methods

Downstream tasks:

- 1. Zero-Shot Image Classification**
 - Classify via similarity of ITM head
- 2. Zero-Shot Image-Text Retrieval**
- 3. Linear probing (last image encoder layer)**

Similarity analysis:

Centered Kernel Alignment (CKA)

- Invariant to isotropic scaling and orthogonal transformations

Datasets (ZS Classification & Linear probe): ImageNet, Caltech101, Caltech256, CIFAR10, CIFAR100, Describable Textures Dataset (DTD), FGVC Aircraft, Food101, Oxford-IIIT Pet, Places365, Stanford Cars, and STL10

Datasets (ZS Retrieval): MS COCO, Flickr8k, Flickr30k

(2) Create dataset classifier from label text

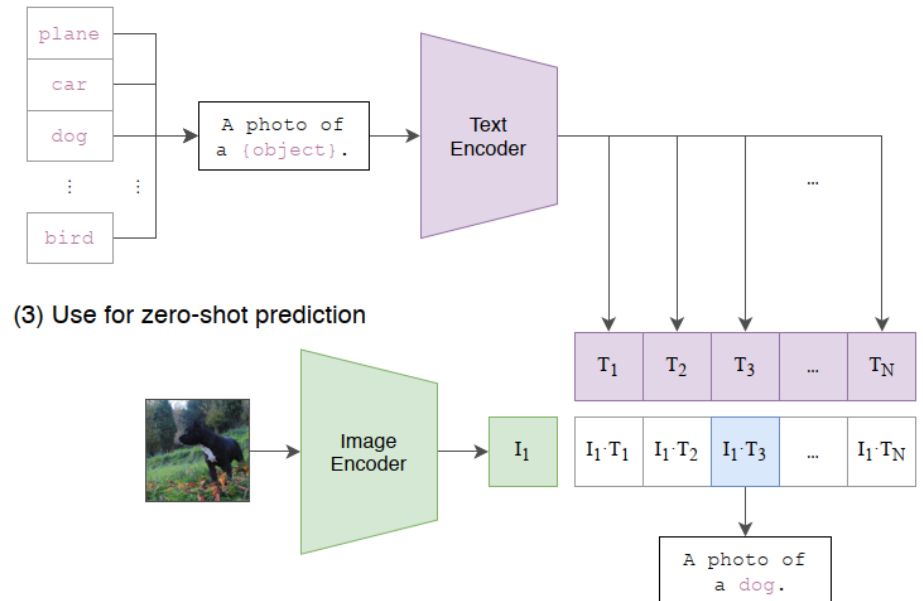


Figure: Zero-shot classification with CLIP

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Related work

Loss ablation studies

- ***SLIP*: CLIP + SimCLR**
 - Addition of SimCLR loss improves over CLIP baseline
- *DeCLIP*: “Data-efficient CLIP”
 - Self-supervision in each modality
 - Multi-view supervision: stochastic augmentations on each modality
 - Nearest-neighbor supervision: sample similar captions for more positive examples
 - Ablation of types of supervision, but not of individual loss functions

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Language supervision studies

- „*Does language help generalization in vision models?*“
 - Compares several multimodal and vision-only models
 - Finds that multimodal models don’t perform better
- „***Is a caption worth a thousand images?***“
 - Controlled study of CLIP vs SimCLR
 - Both losses contrast views of same underlying scene
 - CLIP learns better representations than SimCLR when more data is available

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CLIP learns better *visual* representations than ITM or SimCLR

CLIP outperforms SimCLR and ITM models on linear probing task

- Between 12%-30% better accuracy
- Contrastive loss better than Matching loss
- Language supervision better than image-side self-supervision (SimCLR)

SimCLR learns better visual representations than ITM

Model	Average (All)
CLIP+MLM	67.83
CLIP	67.05
CLIP+ITM	66.96
CLIP+ITM+MLM	66.60
CLIP+SimCLR+ITM	64.60
CLIP+ITM+SimCLR+MLM	65.95
CLIP+SimCLR+MLM	65.38
CLIP+SimCLR	64.59
SimCLR+ITM+MLM	52.81
SimCLR+ITM	43.01
SimCLR	42.71
ITM+MLM	32.09

Table: Average accuracy (Top-1) on linear probe task

CLIP learns better *multimodal* representations than ITM

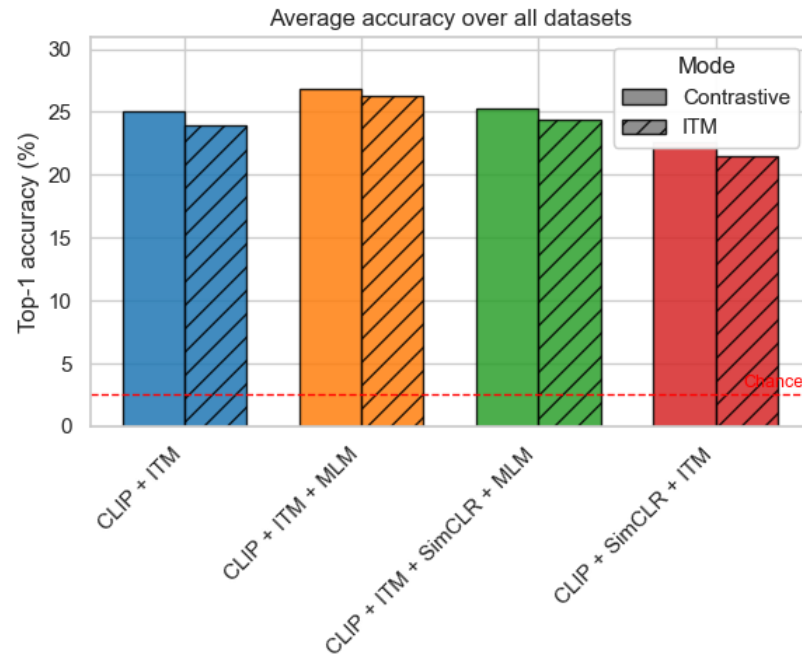


Figure: Zero-shot classification accuracy of CLIP vs ITM (CLIP models)

- Average accuracy **~25%**
- ITM head has worse accuracy than CLIP similarity

CLIP learns better *multimodal* representations than ITM

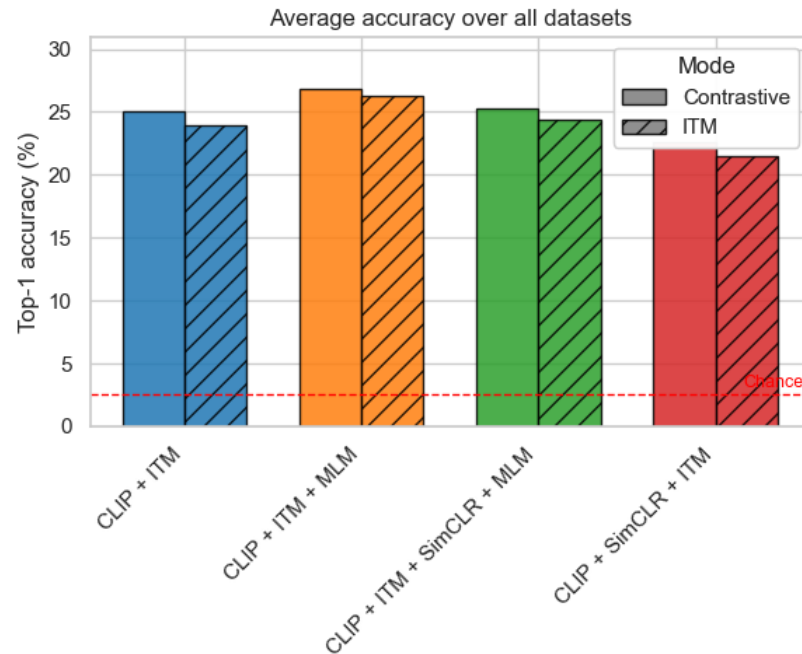


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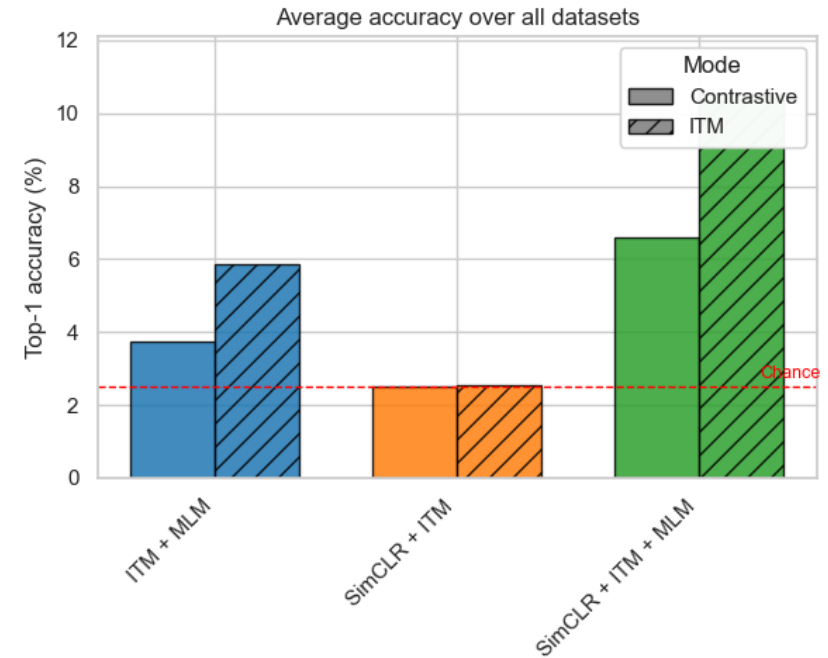


Figure: Zero-shot classification accuracy of CLIP vs ITM (ITM models)

- Average accuracy ~**2-11%**
- ITM head improves accuracy...
- ...but ITM models have much lower performance than CLIP models

MLM improves CLIP, SimCLR & ITM degrade CLIP

Linear probe:

- MLM improves CLIP marginally (~1%)
- SimCLR and ITM worsen accuracy

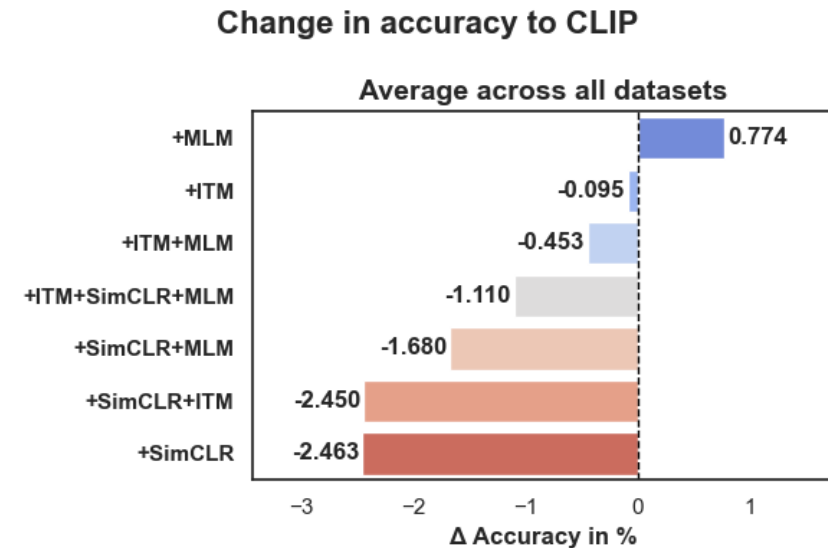


Figure: Change in accuracy for CLIP-based models compared to CLIP baseline

MLM improves CLIP, SimCLR & ITM degrade CLIP

Zero-shot classification:

- **CLIP + MLM** and **CLIP + MLM + ITM** improve over CLIP
- All other models are worse
 - Adding SimCLR or ITM degrades performance

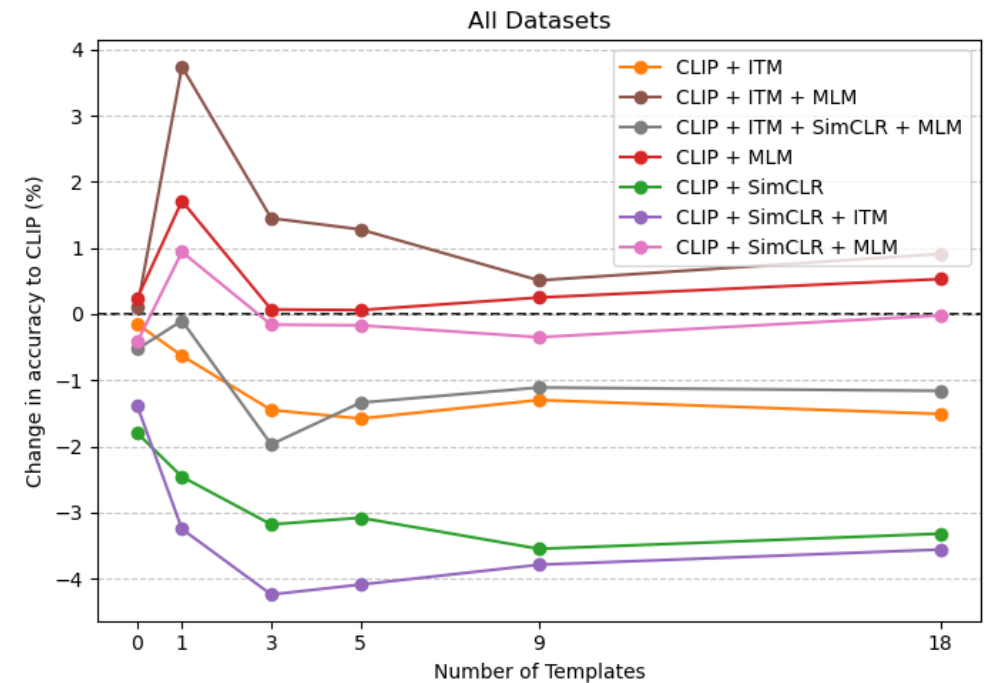


Figure: Change in accuracy for CLIP-based models compared to CLIP baseline, as a function of templates used

MLM improves CLIP, SimCLR & ITM degrade CLIP

Zero-shot retrieval:

- MLM improves CLIP
 - More than in Zero-Shot Classification
 - Increased fine-grained alignment from MLM
- SimCLR and ITM decrease performance

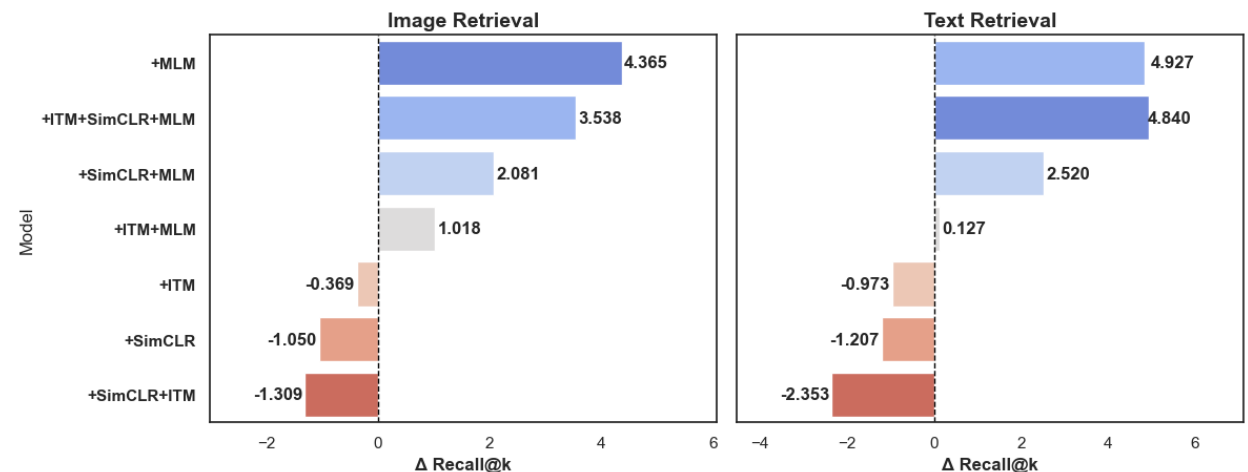


Figure: Change in accuracy for CLIP-based models compared to CLIP baseline

CLIP + unimodal losses improve performance in low-data regime

Notes:

- Models are trained with weaker augmentation settings
- Which has a positive impact on interaction between CLIP and SimCLR
- Discussion will follow

CLIP + unimodal losses improve performance in low-data regime

Zero-shot classification:

- Adding SimCLR or MLM to CLIP improves performance by ~2% below the 40% fraction

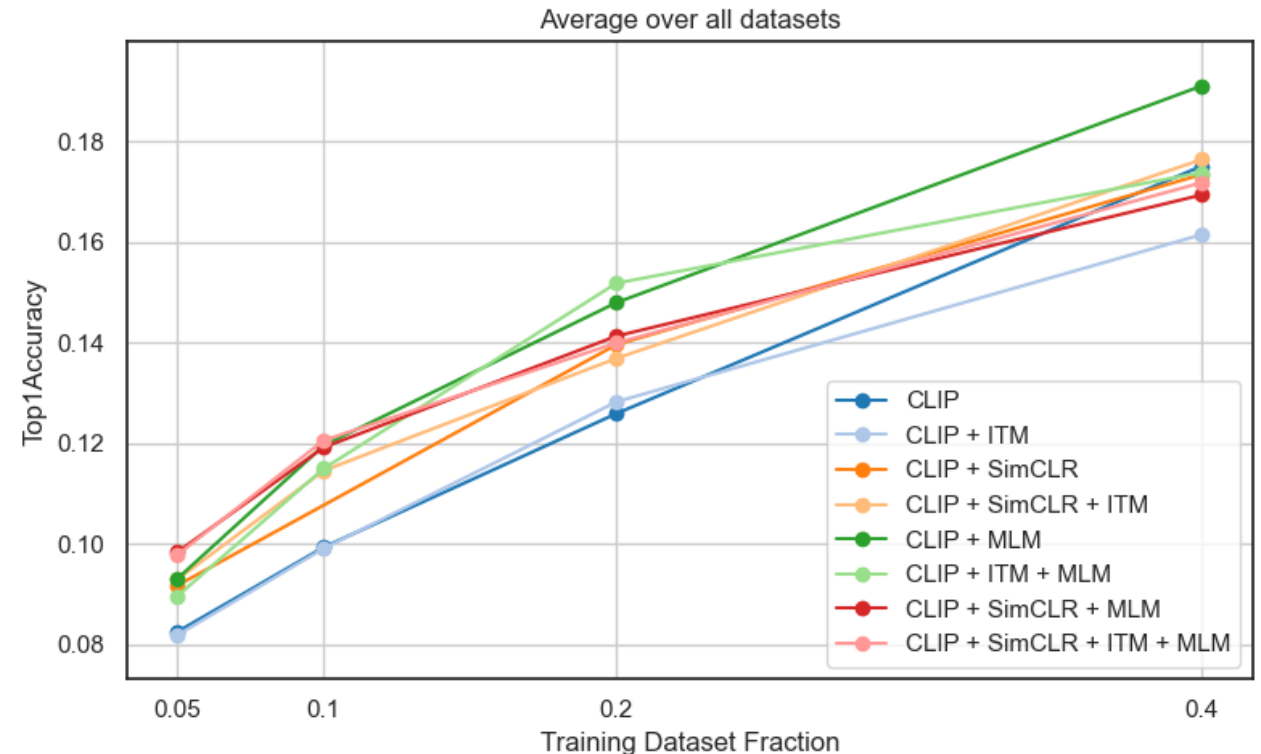


Figure: ZS classification accuracy (averaged) for each model for fractions up to 40%

CLIP + unimodal losses improve performance in low-data regime

Zero-shot retrieval:

- SimCLR and MLM give slightly less benefit in retrieval
- Below 20% fraction

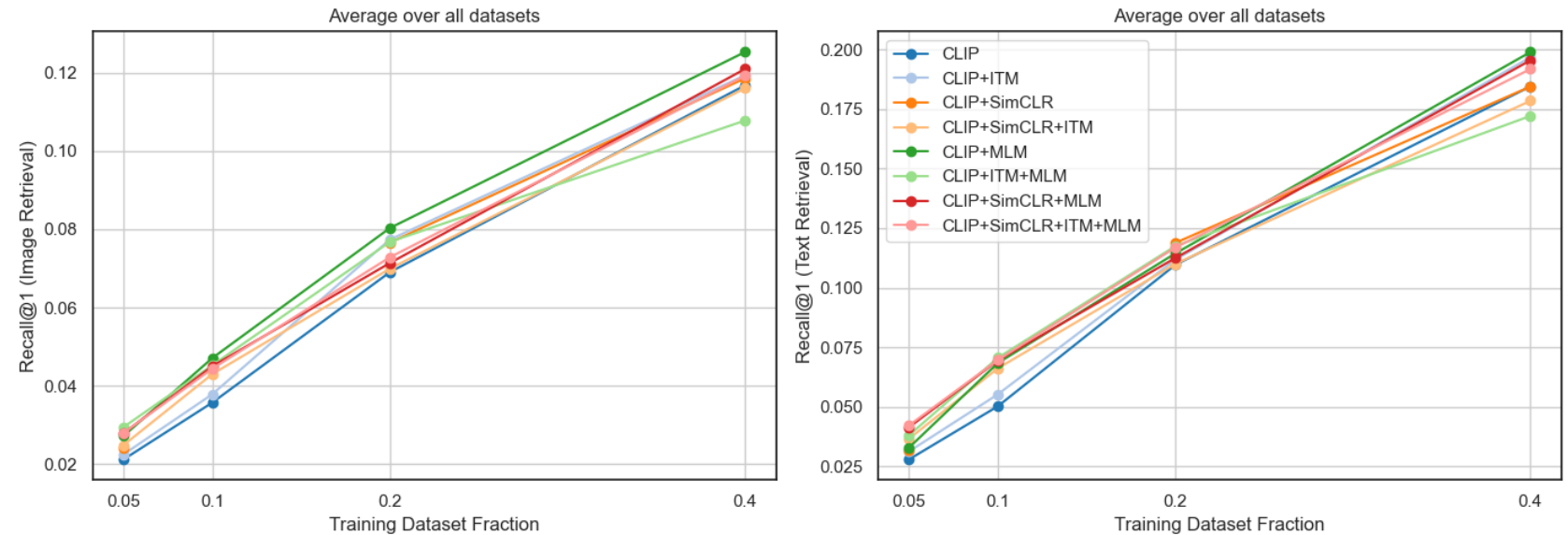


Figure: ZS retrieval recall@1 (averaged) for each model for fractions up to 40%

CLIP + unimodal losses improve performance in low-data regime

Linear probe:

- Pure CLIP shows worse performance below 20% fraction (blue lines)
- Adding unimodal losses improves accuracy
- SimCLR models have best performance at 5% fraction (violet lines)

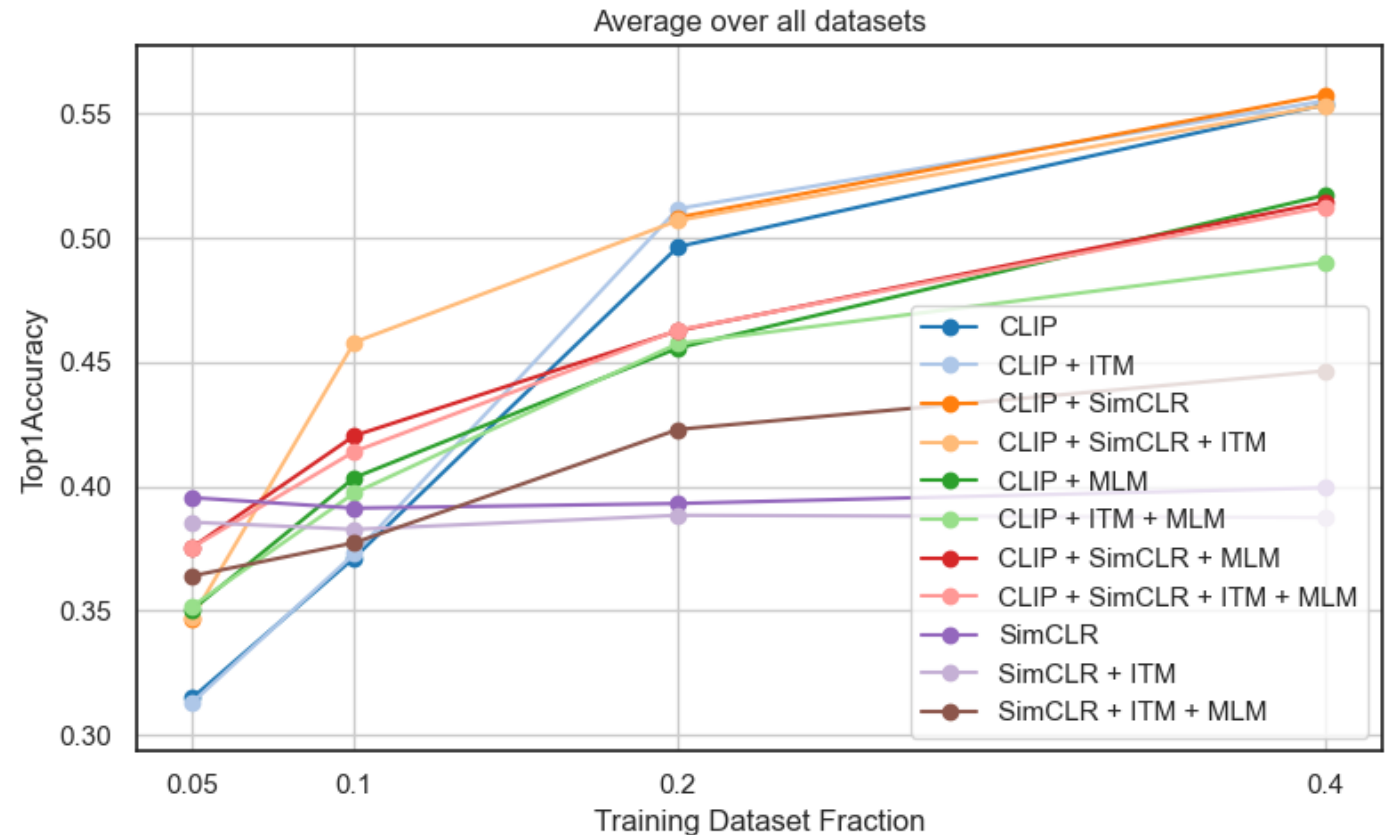
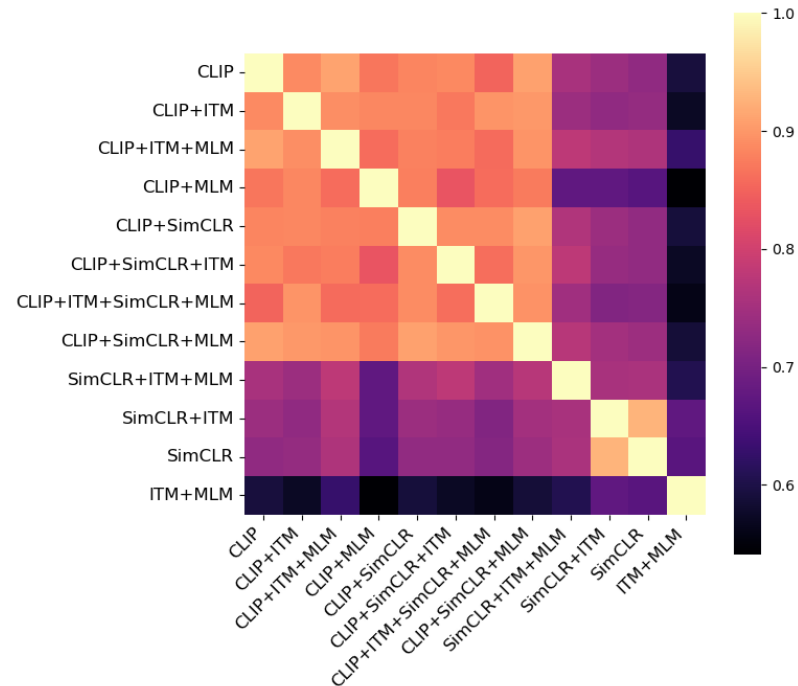


Figure: Linear probe accuracy (averaged) for each model for fractions up to 40%

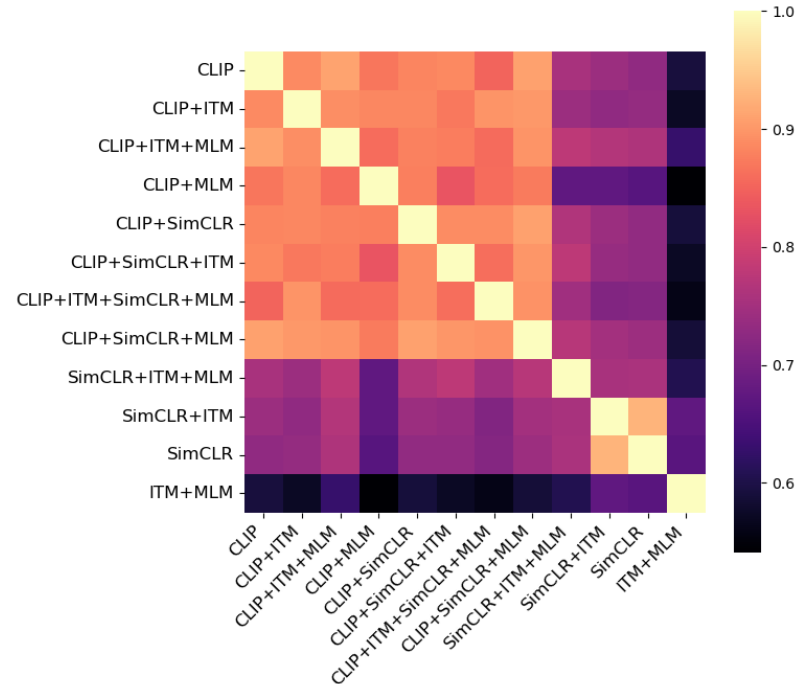
Similarity of model representations (CIFAR-10 dataset)



Last encoder layer

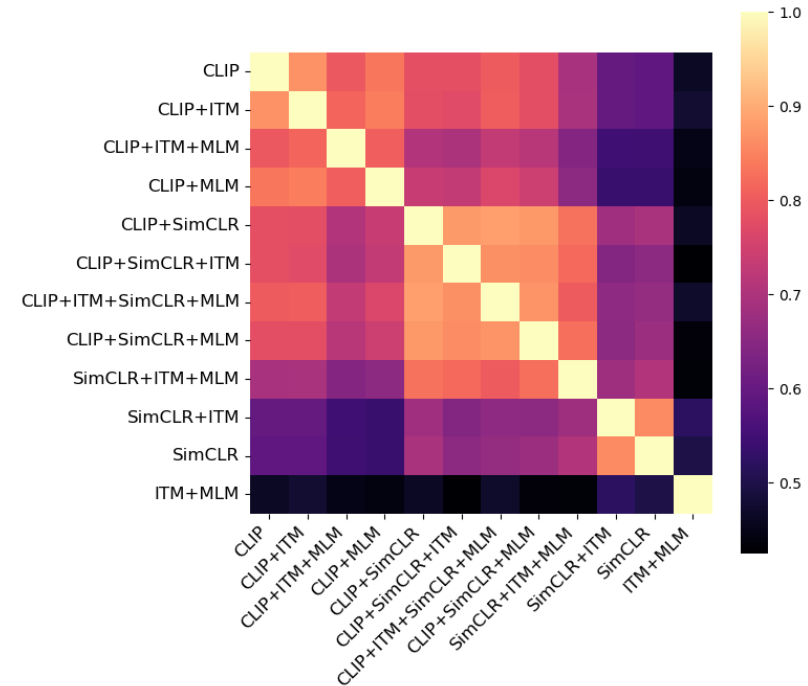
- CLIP dominates representational similarity
- SimCLR shows lesser influence
- ITM shows no influence on representations
 - ITM models don't cluster

Similarity of model representations (CIFAR-10 dataset)



Last encoder layer

- CLIP dominates representational similarity
- SimCLR shows lesser influence
- ITM shows no influence on representations
 - ITM models don't cluster



Visual projection layer

- SimCLR has stronger influence on representations
 - Loss applied at this layer
- ITM still shows no influence

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Discussion of loss functions

CLIP

- Produces best models
 - Better than ITM for learning multimodal representations
 - Better than ITM and SimCLR for learning visual representations
- But needs more data to learn good representations
 - Unimodal losses can help in low-data regime

ITM

- Not sufficient to learn good visual or multimodal representations
- No extensive cross-modal modelling like in other implementations
 - Limited by use of CLIP loss

Discussion of loss functions

MLM

- Always increases performance when added to CLIP
- Additional training of text stream more beneficial than visual stream?
 - Paper: “*LiT: Locked-image Text Tuning*”

SimCLR

- Degrades performance of CLIP
- Interferes with CLIP loss at projection layer

Discussion of loss functions

MLM

- Always increases performance when added to CLIP
- Additional training of text stream more beneficial than visual stream?
 - Paper: “*LiT: Locked-image Text Tuning*”

SimCLR

- Degrades performance of CLIP
- Interferes with CLIP loss at projection layer
- Stronger augmentations lead to more interference

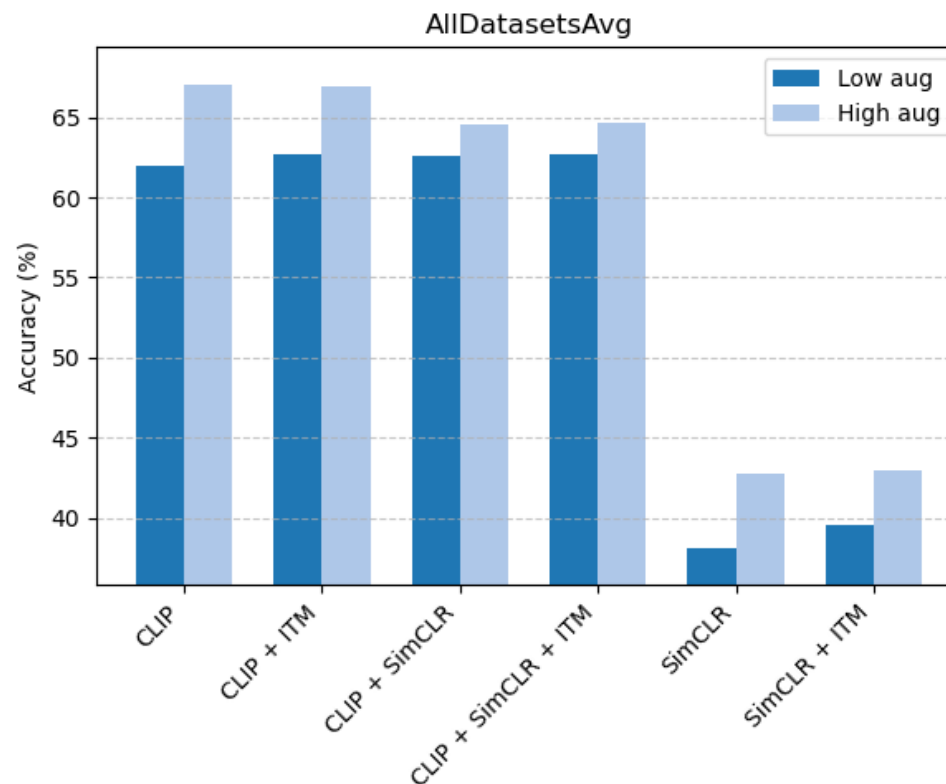


Figure: Linear probe accuracy for different augmentation strengths

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Limitations and Future Work

- Suboptimal implementation of SimCLR
 - Future work: Change to separate projection heads
- Architecture limits loss functions
 - Cross-modal losses are incompatible with CLIP loss
 - Future work: use different architecture that enable other loss functions (single stream or dual-stream with cross-attention layers)
- Limited selection of loss functions
 - Future work: Investigate other losses (Masked Patch Prediction, different ITM implementation, fine-grained CLIP loss, etc.)

Thank you!